

A. Customer Nodes Exploration

--1. How many unique nodes are there on the Data Bank system?

```
select count(distinct node_id) as unique_nodes  
from customer_nodes;
```

	unique_nodes bigint
1	5

--2. What is the number of nodes per region?

```
select r.region_id,region_name,count(node_id) as node_count  
from regions r  
left join customer_nodes cn  
on r.region_id = cn.region_id  
group by r.region_id,region_name;
```

	region_id integer	region_name character varying (9)	node_count bigint
1	1	Australia	770
2	2	America	735
3	5	Europe	616
4	4	Asia	665
5	3	Africa	714

--3. How many customers are allocated to each region?

```
select r.region_id,region_name,count(customer_id) as customer_count
from regions r
left join customer_nodes cn
on r.region_id = cn.region_id
group by r.region_id,region_name;
```

	region_id integer	region_name character varying (9)	customer_count bigint
1	1	Australia	770
2	2	America	735
3	5	Europe	616
4	4	Asia	665
5	3	Africa	714

--4. How many days on average are customers reallocated to a different node?

```
with day_tbl as (
    select *,(end_date - start_date) as days
    from customer_nodes
    where end_date <> '9999-12-31'
    order by customer_id
)
,days as (
    select customer_id,node_id,SUM(days) as days_summed
    from day_tbl dt
    group by customer_id,node_id
)
select ROUND(AVG(days_summed)) as average_days_before_node_gets_relocated
from days;
```

	average_days_before_node_gets_relocated numeric
1	24

```
--5. What is the median, 80th and 95th percentile for this same
-- reallocation days metric for each region?
with day_tbl as (
  select *,(end_date - start_date) as days
  from customer_nodes
  where end_date <> '9999-12-31'
  order by customer_id
)
,region_tbl as (
  select customer_id,region_id,node_id,SUM(days) as days_summed
  from day_tbl dt
  group by customer_id,region_id,node_id
)
select region_id
  ,ROUND(percentile_cont(0.5) WITHIN GROUP (ORDER BY days_summed)) as median
  ,ROUND(percentile_cont(0.8) WITHIN GROUP (ORDER BY days_summed)) as p80
  ,ROUND(percentile_cont(0.95) WITHIN GROUP (ORDER BY days_summed)) as p95
from region_tbl
group by region_id;
```

	region_id integer	median double precision	p80 double precision	p95 double precision
1	1	21	34	51
2	2	22	34	54
3	3	22	35	54
4	4	22	35	52
5	5	23	34	51

B. Customer Transactions

```
--1. What is the unique count and total amount for each transaction type?

select txn_type,count(distinct customer_id) as unique_count_of_transaction_type,sum(txn_amount) as amount_sum
from customer_transactions
group by txn_type

--2. What is the average total historical deposit counts and amounts for all customers?
with temp as (
  select customer_id,count(*) as cnt_dep,AVG(txn_amount) as sum_amt
  from customer_transactions
  where txn_type = 'deposit'
  group by customer_id
)
select ROUND(AVG(cnt_dep)) as avg_dep_counts,ROUND(AVG(sum_amt)) as avg_dep_amount
from temp;
```

	avg_dep_counts numeric	avg_dep_amount numeric
1	5	509

--3. For each month - how many Data Bank customers make more than 1 deposit
-- and either 1 purchase or 1 withdrawal in a single month?

```
with month_tbl as (  
    select *,EXTRACT(month from txn_date) as txn_month  
    from customer_transactions  
)  
,cnt as (  
    select txn_month,customer_id,txn_type,count(*) as cnt_txn_type  
    from month_tbl  
    group by txn_month,customer_id,txn_type  
    order by txn_month,customer_id  
)  
,pivotted_tbl as (  
    select txn_month,customer_id,  
        SUM( CASE when txn_type = 'deposit' then cnt_txn_type else 0 END) as deposit,  
        SUM( CASE when txn_type = 'purchase' then cnt_txn_type else 0 END) as purchase,  
        SUM( CASE when txn_type = 'withdrawal' then cnt_txn_type else 0 END) as withdrawal  
    from cnt  
    group by txn_month,customer_id  
)  
select txn_month,count(*) as customer_cnt  
from pivotted_tbl  
where deposit > 1 and (purchase >= 1 or withdrawal >= 1)  
group by txn_month
```

	txn_month numeric	customer_cnt bigint
1	1	168
2	2	181
3	3	192
4	4	70

--4. What is the closing balance for each customer at the end of the month?

```

with month_tbl as (
  select *,EXTRACT(month from txn_date) as txn_month
  from customer_transactions
)
,amount_tbl as (
  select txn_month,customer_id,txn_type,sum(txn_amount) as sum_txn_amount
  from month_tbl
  group by txn_month,customer_id,txn_type
  order by txn_month,customer_id
)
,pivot_tbl as (
  select txn_month,customer_id,
    SUM(CASE when txn_type = 'deposit' then sum_txn_amount else 0 END) as deposit,
    SUM(CASE when txn_type = 'purchase' then sum_txn_amount else 0 END) as purchase,
    SUM(CASE when txn_type = 'withdrawal' then sum_txn_amount else 0 END) as withdrawal
  from amount_tbl
  group by txn_month,customer_id
  order by customer_id,txn_month
)
,m_bal as (
  select *,(deposit - purchase - withdrawal) as monthly_bal
  from pivot_tbl
)
select *,SUM(monthly_bal) OVER(PARTITION BY customer_id ORDER BY txn_month RANGE between unbounded preceding and current row) as net_balance
from m_bal;

```

	txn_month numeric	customer_id integer	deposit numeric	purchase numeric	withdrawal numeric	monthly_bal numeric	net_balance numeric
1	1	1	312	0	0	312	312
2	3	1	324	1276	0	-952	-640
3	1	2	549	0	0	549	549
4	3	2	61	0	0	61	610
5	1	3	144	0	0	144	144
6	2	3	0	965	0	-965	-821
7	3	3	0	0	401	-401	-1222
8	4	3	493	0	0	493	-729
9	1	4	848	0	0	848	848
10	3	4	0	193	0	-193	655
11	1	5	1780	0	826	954	954
12	3	5	1130	2521	1486	-2877	-1923
13	4	5	0	0	490	-490	-2413
14	1	6	1627	894	0	733	733
15	2	6	346	962	169	-785	-52
16	3	6	2749	968	1389	392	340
17	1	7	964	0	0	964	964
18	2	7	2286	77	0	2209	3173
19	3	7	723	1290	73	-640	2533
20	4	7	615	0	525	90	2623

```
--5. What is the percentage of customers who increase their closing balance by more than 5%?
with month_tbl as (
    select *,EXTRACT(month from txn_date) as txn_month
    from customer_transactions
),
amount_tbl as (
    select txn_month,customer_id,txn_type,sum(txn_amount) as sum_txn_amount
    from month_tbl
    group by txn_month,customer_id,txn_type
    order by txn_month,customer_id
),
pivot_tbl as (
    select txn_month,customer_id,
        SUM(CASE when txn_type = 'deposit' then sum_txn_amount else 0 END) as deposit,
        SUM(CASE when txn_type = 'purchase' then sum_txn_amount else 0 END) as purchase,
        SUM(CASE when txn_type = 'withdrawal' then sum_txn_amount else 0 END) as withdrawal
    from amount_tbl
    group by txn_month,customer_id
    order by customer_id,txn_month
),
m_bal as (
    select *,(deposit - purchase - withdrawal) as monthly_bal
    from pivot_tbl
),
lead_tbl as (
    select *,LEAD(monthly_bal) OVER(PARTITION BY customer_id ORDER BY txn_month) as leading_bal from m_bal
),
perc_tbl as (
    select *,ROUND((100 * (leading_bal - monthly_bal)/abs(monthly_bal)),2) as percentage_change
    from lead_tbl
    where leading_bal is not null
)
select ROUND(100*count(distinct customer_id)/(select count(distinct customer_id) from customer_transactions),2) as percentage_of_customers
from perc_tbl
where percentage_change >= 5;
```

	percentage_of_customers 
1	79.00